

D1 Justifications of my design choices

Advantages and disadvantages of my app based on my design

Use case diagram Advantages of proposed solution

- Automation of feedback –An ai automatically reviews the recording so user know they are pronouncing words correctly and the ai means the teachers don't have to do it. (practise game)
- Operational efficiency – by saving users settings within their accounts, it reduces the amount of unnecessary effort having to re enter settings and all functions are streamlined so the app operates smoothly as possible (homepage and flow charts)
- Cost-effectiveness – other than the future ai(see M3) there are minimal creation costs and so it is very cost effective as there is no consistent cost and only a relatively large initial cost for the server and labour
- Globalisation – being able to select their native language and their learning language means that anyone anywhere would eventually be able to use it and as the logo is clearly shown on the app it will give the school international influence (homepage)
- Improved communication- the speed game will help the user get used to making links between their native language and the language they're learning so this can help them better communicate in conversation on the fly and the practise game will help them improve their pronunciation (speed and practise)
- Customisation and adaptability – the difficulty and learning language selection allows the user to select how they experience the app and their learning (homepage)
- Increased markets – as more languages and modules are added it enables better communication for all users allowing them to gain opportunities in new markets (homepage)
- Ease of access for customers - easy to access all of the functions of the application as most are available to access from and visible on the main page (homepage)
- New marketing opportunities – native and learning language is in drop down menus so more languages can easily be added allowing the app to reach markets across the world
- Customer or user information – records use information and activity information but its eventual security has not been planned yet and is based on the quality of teachers passwords (login, homepage and dfd)
- Real-time information – takes in users audio within the practise game which can be given as output or submitted for review (practise game)
- New employment – gives lots of opportunities for new employment as you can learn a language which is very valuable (although not many people would accept a qualification based on an app)

Disadvantages of proposed solution

- Financial cost –this ai will have a very high cost so should only be invested in if the app is successful (practise game)
- Changeover risks- other than the development of the app the changeover has no costs as it will be running alongside the in class teaching the biggest risk is that the app is not popular and doesn't take off as intended because that would mean the app was a waste of money
- Training- needs the app does have many functions so it could be confusing to staff so if required training will be given but won't take long or cost much but to understand how the data flows would almost definitely require training (Data flow diagram)
- Lack of job security and job losses – may lower the job security for language teachers as a whole if it is successful and useful enough but not to a major degree
- Security issues- based on how basic teachers passwords are it is possible their accounts will be hacked and someone will misuse any student information stored within although it is unlikely it is not impossible (login)
- Privacy issues - the users may consider the large amount of use data that is kept as a privacy issue as although more information is helpful for the teachers it is not as good for the students as no one likes their (dfd)
- Potential customer concerns- security of their data and how well what they learn can be translated into working lives or just their general lives (login)
- Loss of personal contact- the app in general has a bit of an impersonal touch as it is a computer game so it feels a lot more impersonal than being taught in person by a teacher/tutor so this may lower some peoples use of the app